Lab Work 4: Extraction, Cleansing, Loading and Transformation of Excel Data from Website in Local SQL Server in Monthly Basis

1. Download the Excel File

Visit the provided URL and manually download the Excel file for the latest monthly statistics.

https://www.nrb.org.np/category/monthly-statistics/?department=bfr

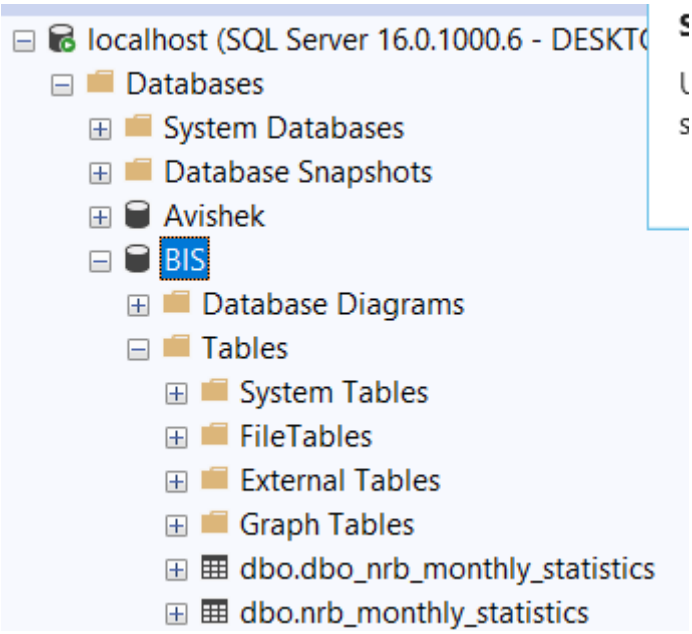
2. Install Required Python Libraries

Ensure the following libraries are installed in your environment:

```
pip install pandas sqlalchemy pyodbc openpyxl
```

3. Key Points in the Code

1. **Database Setup**:
   - Ensure the database BIS and table (`MonthlyStatistics`) exist in SQL Server.
   - The table schema must match the structure of the Excel file and a column name should be added as the name of the month and year same as in name of the excel file



2. **Excel Sheet Processing**:
   - The script reads all sheets from the Excel file.

- Each sheet's data is appended to the same SQL table every month. For now refer to sheet C14 and data **District wise List of Banks and Financial Institutions Branches.**

1. Data Extraction: The monthly Excel file is downloaded manually from the Nepal Rastra Bank official website. The dataset contains multiple sheets, but for this lab work, the focus is on Sheet C14: District-wise List of Banks and Financial Institutions Branches. Python's pandas library is used to read the Excel file and load the content into a DataFrame.

2. Data Cleansing: The raw dataset often contains unnecessary rows such as headers, footers, and blank spaces, which are removed during cleansing. Columns are renamed so they match the SQL Server table schema. Numeric values that were initially stored as text are converted to proper numerical data types. Null or missing values are checked and either filled with zero or dropped. An additional column is added to include the Month_Year, extracted from the Excel filename, to identify the reporting period.

3. Database Setup: A database named BIS is created in SQL Server, and a table called MonthlyStatistics is defined with the following fields: Province, District, Class_A, Class_B, Class_C, Total, Population, Pop_Per_Branch, and Month_Year. This schema ensures that each month's data can be appended consistently.

4. Data Loading: Using SQLAlchemy and pyodbc, a connection is established between Python and SQL Server. The cleansed DataFrame is appended into the MonthlyStatistics table. Each new monthly dataset is stored with its Month_Year tag, ensuring that historical records are maintained for future analysis.

5. Data Transformation: Derived values such as Pop_Per_Branch are calculated and inserted along with the dataset. This enables later reporting and aggregation tasks, such as comparing branch distribution across provinces and analyzing growth trends.

6. Automation (Future Scope): Although this lab work demonstrates a semi-manual process, the workflow can be automated by scheduling the Python script. By dynamically reading the latest file and appending data into SQL Server, the system can be scaled into a fully automated monthly ETL pipeline.

The overall process followed in this lab work can be divided into five major stages: Extraction, Cleansing, Database Setup, Loading, and Transformation.

Process

SQL Queries

1. Table Creation

```
CREATE DATABASE BIS;
GO
USE BIS;
GO
CREATE TABLE MonthlyStatistics (
    Province NVARCHAR(100),
    District NVARCHAR(100),
    Class_A INT,
    Class_B INT,
    Class_C INT,
    Total INT,
    Population BIGINT,
    Pop_Per_Branch DECIMAL(18,2),
    Month_Year NVARCHAR(20)
);
```

2. Insertion Example (from cleaned Python DataFrame):

```
INSERT INTO MonthlyStatistics
(Province, District, Class_A, Class_B, Class_C, Total, Population, Pop_Per_Branch, Month_Year)
VALUES
('Bagmati', 'Kathmandu', 50, 30, 20, 100, 1000000, 10000.00, 'July-2024');
```

(In practice, the Python script inserts all rows dynamically using SQLAlchemy's to_sql method, so manual inserts are only for demonstration.)

Learning Outcomes

1. **Data Integration**: Extract, clean, and upload Excel data into SQL Server efficiently using Python.
2. **Automation & Scheduling**: Automate monthly data import tasks with dynamic scripting and scheduling tools.
3. **ETL Proficiency**: Develop foundational ETL skills for structured data handling, storage, and analysis.

	StatID	StatMonth	Province	District	Class_A	Class_B	Class_C	Total	Population	Population_Per_Branch
1	1	2025-07-01	Koshi	Taplejung	24	0	0	24	120590	5024.58
2	2	2025-07-01	Koshi	Panchthar	28	3	2	33	172400	5224.24
3	3	2025-07-01	Koshi	Ilam	38	9	1	48	279534	5823.63
4	4	2025-07-01	Koshi	Jhapa	139	53	12	204	998054	4892.42
5	5	2025-07-01	Koshi	Sankhuwasabha	32	1	0	33	158041	4789.12
6	6	2025-07-01	Koshi	Bhojpur	24	0	1	25	157923	6316.92
7	7	2025-07-01	Koshi	Terhathum	21	3	0	24	88731	3697.13
8	8	2025-07-01	Koshi	Dhankuta	28	9	1	38	150599	3963.13
9	9	2025-07-01	Koshi	Morang	182	61	5	248	1148156	4629.66
10	10	2025-07-01	Koshi	Sunsari	133	39	13	185	926962	5010.61
11	11	2025-07-01	Koshi	Solukhumbu	28	0	0	28	104851	3744.68